Consider a vocabulary $\mathcal{V}$,

$\mathcal{V}$ =([start], building, character, a, is, astronomy, science, experience, natural,[end]).

Assume that we have a pre-trained GPT model for text generation and the first input token to the model is always the special token [start]. The prediction probabilities outputted by the model are given below.

$$\hat{Y} = \begin{matrix} t_0 \\ t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \end{matrix} \begin{bmatrix} 0.05 & 0.02 & 0.14 & 0.07 & 0.09 & 0.41 & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & 0.43 & 0.01 & 0.06 & 0.09 & 0.01 & 0.04 \\ 0.05 & 0.07 & 0.28 & 0.29 & 0.03 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & 0.48 & 0.02 & 0.01 & 0.07 & 0.06 & 0.14 & 0.07 & 0.01 \\ 0.08 & 0.29 & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.22 & 0.13 & 0.08 & 0.01 & 0.03 & 0.06 & 0.23 & 0.08 & 0.03 \end{bmatrix}$$

The zeroth row of the matrix is the output probability distribution by the model given the special [Start] token as input. Following the Greedy Search decoding strategy, the subsequent rows give the conditional probability distribution conditioned over the previous tokens. If you think the given information is insufficient for any sub-questions, then enter -1 as your answer

1) What is the probability of the predicted word equal to ``Astronomy" given the ``[Start]" token as the initial input $P(y_1 = Astronomy | y_0 = [Start]$?

$\mathcal{V}$ =([start], building, character, a, is, astronomy, science, experience, natural,[end]).

ssume that we have a pre-trained GPT model for text generation and the first input token to the model is always the special token [start]. Tl

$$\begin{bmatrix} 0.05 & 0.02 & 0.14 & 0.07 & 0.09 & 0.41 & 0.08 & 0.05 & 0.08 & 0.01 \end{bmatrix}$$

$\therefore$ $P(y_1 = \text{Astronomy} \mid y_0 = [\text{Start}]) = 0.41$

2) What is the probability of the sequence ``Astronomy is a character building experience"?

P (Astronomy is a character building experience)

$= P(\text{Astronomy} \mid \text{start}) \cdot P(\text{is} \mid \text{Astronomy, start}) \cdot P(\text{a} \mid \text{is, Astronomy, start})$

$\cdot P(\text{character} \mid \text{a, is, astronomy, start}) \cdot P(\text{building} \mid \text{ch, a, is, astr., start}) \cdot$

$\cdot P(\text{experience} \mid \text{bldg, ch, a, is, Astronomy}) \cdot P(\text{end} \mid \text{start, Astr, is, a, character, bldg, ex})$

$= 0.41 \times 0.43 \times 0.29 \times 0.48 \times 0.29 \times 0.23 = 0.001636882032$

However in absence of $P(\text{end} \mid \ldots)$, answer can be -1

3) What is the probability of the sequence `` Astronomy is a natural science "?

$= P(\text{A} \mid \text{start}) \cdot P(\text{is} \mid \text{A, start}) \cdot P(\text{a} \mid \text{is, A, start}) \cdot P(\text{natural} \mid \text{a, is, A, st.}) \cdot P(\text{Science} \mid \text{natural})$

$0.41 \times 0.43 \times 0.29 \times ? \times ?$

But this follows greedy search.

$\therefore$ at time step 4, it will feed 'a' as $P(\text{natural}) = 0.07$

for next time step it will feed. Character not natural

∴ answer should be −1

4) Suppose we use Top−k sampling, with $k = 3$, What is the probability of the predicted word equal to ``Astronomy" given the ``[Start]" token as the initial input $P(y_1 = Astronomy | y_0 = [Start])$?

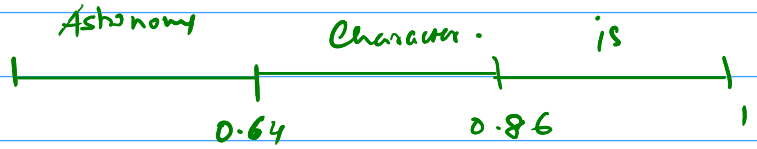
$\mathcal{V}$ =([start], building, character, a, is, astronomy, science, experience, natural,[end]).

ssume that we have a pre-trained GPT model for text generation and the first input token to the model is always the special token [start]. Th

$$[0.05 \quad 0.02 \quad 0.14 \quad 0.07 \quad 0.09 \quad 0.41 \quad 0.08 \quad 0.05 \quad 0.08 \quad 0.01]$$

top 3 K = Astronomy (0.41)
character (0.14)
is (0.09)

Astronomy $= \frac{0.41}{0.41+0.14+0.09} = 0.64$, $P(\text{character}) = \frac{0.14}{0.64} = 0.22$, $P(\text{is}) = 0.14$



Now we find Random (0,1) & select as per mat.
In question random integer not given ∴ answer is −1

5) Let us consider the vocabulary $\mathcal{V}$ given below.

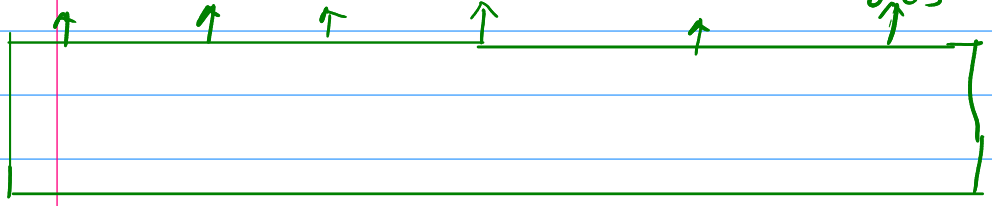
$\mathcal{V}$ =([CLS], building, character, a, is, astronomy, science, experience, natural,[SEP]).

Now we train the BERT model by randomly initializing the parameters. Suppose we feed in the following input sequence "Astronomy is a [mask] building [Mask] ", where we replaced the words ``character" and ``experience" with the special [Mask] token. We just feed this single sentence (segment) to the model and do Masked Language Modelling (MLP). Note that the special tokens [CLS] and [SEP] were not used. The matrix below gives the prediction probabilities for the given input sequence

$$\hat{Y} = \begin{bmatrix} 0.05 & 0.02 & 0.41 & 0.07 & 0.09 & 0.14 & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & 0.04 & 0.01 & 0.06 & 0.09 & 0.01 & 0.43 \\ 0.05 & 0.07 & 0.28 & 0.03 & 0.29 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & 0.14 & 0.02 & 0.01 & 0.07 & 0.06 & 0.48 & 0.07 & 0.01 \\ 0.08 & 0.29 & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.13 & 0.22 & 0.08 & 0.01 & 0.03 & 0.06 & 0.23 & 0.08 & 0.03 \end{bmatrix}$$

character : 0.10, 0.02, 0.14, 0.02, 0.01, 0.07, 0.06, 0.48, 0.07, 0.01

0.14, 0.13, 0.22, 0.08, 0.01, 0.03, 0.06, 0.23, 0.08, 0.03



Astronomy is a <mask> building <mask>
character experience.

What is the total loss value incurred by the model for the given input sequence? Use natural logarithm where required

$= -\sum \ln \hat{y} = -\ln(0.14) - \ln(0.23) = 1.966 + 1.469$
$= \underline{3.435}$

6) Suppose we use the pre-trained BERT model for a sentiment classification problem. Assume that we use the model for feature extraction. That is, the model gives the representation for the given input sentence and we use that as input to some traditional classification algorithms. Then choose which of the following statements is (are) True

- ✓ We can only take the representation of [CLS] token as an input to the classifier
- We can also use the representation of [SEP] token as an input to the classifier
- We can take representation of any word in the input sequence as an input to the classifier

Intuition

As representation of any word "contain the semantic relations between all words, we can take representation of any word'.

Research :

Great question! Let's break this down to understand why the **[CLS] token** is often used for feature extraction in BERT, even though every token representation contains contextual information.

How Representations in BERT Work:

1. **Contextualized embeddings:**
   BERT generates **contextual embeddings** for every token. This means that each token's representation captures not just the meaning of that token but also information about its surrounding tokens in the input sequence.
2. **[CLS] Token Aggregation:**
   - The **[CLS] token** is a special token added at the beginning of every input sentence.
   - BERT is trained such that the **final hidden state of the [CLS] token** serves as a **summary representation** of the entire sequence, specifically for **classification tasks**.
   - In tasks like sentiment analysis or sentence classification, the [CLS] token is intended to aggregate meaningful information from all tokens, making it a concise feature for input to classifiers.

Why Not Use Any Token Representation for Classification?

While **every token's representation is contextualized** and contains some information about the entire sentence, there are key differences:

1. **Primary Focus vs. Aggregated Information:**
   - Each token's representation focuses **more on itself** and the **local context** (i.e., its meaning in relation to nearby tokens).
   - In contrast, the **[CLS] token** is designed to act as a **global summary** of the entire input, aggregating information across all words.
2. **Task-Specific Training with [CLS]:**
   - When BERT is fine-tuned on classification tasks, it is explicitly optimized to put relevant **classification features** into the [CLS] representation.
   - Other tokens may not be as directly optimized for classification purposes, making their use less effective for sentence-level tasks.
3. **Use Cases for Word Representations:**
   - Individual word representations are more suitable for **token-level tasks** (e.g., named entity recognition, part-of-speech tagging) because these tasks require understanding of individual words in context, not the entire sentence.

∴ Option (a) is correct answer.